Lec33

01 April 2026 02:57 PM

	0	1	2	3
0	1	2	3	4
1	5	6	7	8
2	9	10	11	12
3	13	14	15	16

1 5 9 13 14 10 6 2 3 7 11 15 16 12 8 4

```
for (col = 0; col < arr[0].length; col++){
    if(col%2==0){
        for(row = 0; row < arr.length; row++){
            sout(arr[row][col]
        }
    }
    else{
        for(row = arr.length-1; row >= 0; row--){
            sout (arr[row][col]
        }
    }
}
```

	0	1	2	3
0	1	2	3	4
1	5	6	7	8
2	9	10	11	12
3	13	14	15	16

	0	1	2	3
0	1	5	9	13
1	2	6	10	14
2	3	7	11	15
3	4	8	12	16

01 April 2026 02:57 PM

0 → (0-1) ⟷ (1-0), (0-2) ⟷ (2-0), (0-3) ⟷ (3-0)

1 → (1-2) ⟷ (2-1), (1-3) ⟷ (3-1)

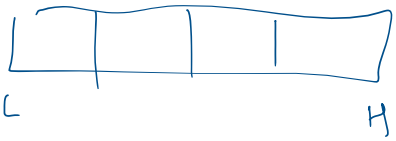
2 → (2-3) ⟷ (3-2)

3 → (3-4)

```
for(row = 0; row < arr.length; row++){
    for(col = row+1; col < arr[0].length; col++){
        temp = arr[row][col]
        arr[row][col] = arr[col][row]
        arr[col][row] = temp
    }
}
```

1	4	7	11	15
2	5	8	12	19
3	6	9	16	22
10	13	14	17	24
18	21	23	26	30

target = 3 / 19



arr[row][col]

```
row = 0
col = arr[0].length-1
while (col>=0 && row<arr.length) {
    if (arr[row][col] == target) {
        return true.
    }
    else if (arr[row][col] > target){
        col--
    }
    else {
        row++
    }
}
return false
```